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Description

Describe your program. What do you envision your app will do?

Note: At this step, do not be concerned about how to do it, just focus on what you envision.

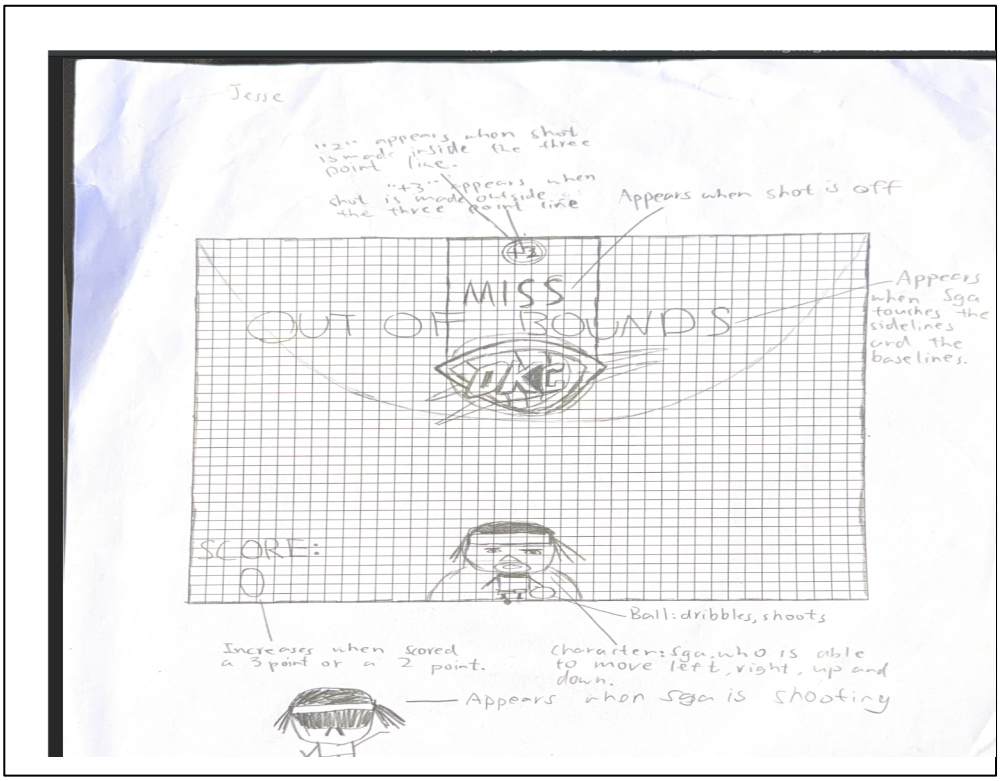
The program that I created was a character name sga, who is able to move right, left, up, and down freely while dribbling on the basketball court, which is the canvas. Sga can take shots either a three pointer or a two pointer, it depends where sga is on the court, whether it is inside the three-point line or outside, there will also be a sign in the hoop saying either “+2” or “+3”. However, he must take the shot without being out of bounds, or else there will be a warning saying, “OUT OF BOUNDS”, and sga is not able to shoot. Sga can take a shot by pressing the space bar, and he’ll turn around to the shoot, so we will see his back. Sga doesn’t always make his shots, the chance of the ball going in the hoop is 60 percent. If he misses, there will be a sign saying, “MISS”. There is also a scoreboard recording the score, adding 0, 2, or 3 points.

Canvas Design

Sketch your canvas. Label any animation or user interaction.

- **Colors:** What colors will be dominant in your app? Are there specific shades that are important to the app's functionality or aesthetic?
- **Animations/Movements:** Will there be any animated elements? If yes, describe how you envision them. For example, will a shape move across the screen?

(0,0)



(400,400)

Code Design

Plan your code. List here the coding concepts and properties you plan to use to accomplish your goals, and why you plan to use them. These should incorporate at least two new concepts from this unit. Briefly describe what purpose each element will serve in your project. Aim for at least four to six properties.

Concept	Purpose
onKeyPress	To move sga with arrow keys and shoot with spacebar.
onStep	To animate ball movement during shooting and driblling.
hitsShape	To see if sga is out of bounds.
distance	To calculate the distance for 2 pointer and a 3 pointer
Random.randint	To randomize shot making.

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Creative Task Reflection

After completing your creative task use this form to reflect on your work.

Difficulties and Opportunities

For this section, go in-depth into either a difficulty you faced or an opportunity you saw to improve or add to your code.

If you Faced a Difficulty:

- Describe the Problem: What specific issue did you encounter?
- Attempted Solutions: What did you try in order to solve the problem?
- Final Solution: What was the final solution that fixed the issue?

If you Found an Opportunity:

- Reason for the Addition/Change: What made you consider this opportunity? Was it a challenge or an aha-moment?
- Impact on Design: How did seizing this opportunity impact other aspects of your design?
- Design Trade-offs: What aspects of your original design did you have to adjust or remove to incorporate this new feature?

If you Faced a Difficulty:

- Describe the Problem: What specific issue did you encounter?
  An issue that I encountered was the one where I didn't what was the best way to make the code know that it was a three pointer or a two pointer.
- Attempted Solutions: What did you try in order to solve the problem?
  I wanted to try doing the similar thing like "Cell reception", which is one of the exercises we had to solve. What it does is if the character reaches a certain height, it activates this. If was doesn't reach the height, it activates another thing.
- Final Solution: What was the final solution that fixed the issue?
  The final solution was the distance of the ball to the hoop. If it was shorter 182, then it's a two pointer, otherwise, it is a 3 pointer.

Finished Project Code Outline

List **only one instance** of each important coding concept used in your project, along with the corresponding line number, and describe how it contributes to the overall function of your program. **Highlight the aspect you are most proud of.** For each concept, also include a brief definition to demonstrate your understanding. By focusing on one instance of each concept, aim to provide more depth and clarity in your responses.

Concept	Line number(s)	Definition	Purpose

onKeyPress	169 - 202	A function that runs whenever a key is pressed.	It moves sga with arrow keys and shoots with space bar.
Group	73 - 106	Shapes can be moved or hide them all at once.	I used it to create sga as one object.
hitsShape	204 - 228	Checks if one shape touches another on the canvas.	I used it to check if sga is out of bounds, and if he is, then sga can't shoot.
angle to & getPointInDir	121 - 125	Finds the direction from one point to another, then moves something using that path.	This makes the ball fly nicely toward the hoop when shooting.

Future goals

Reflect on your entire design and coding process, from conceptualization to implementation. In this section, consider not only what you would change in the final product but also how you would approach the entire project differently.

- **Code Changes:** What specific changes would you make to your code if you had more time or if you were to start from scratch? Would you add new features or refine existing ones?
- **Process Adjustments:** Did you stick to the original plan, or were there deviations? Were these changes beneficial or did they lead to complications? How could your planning or approach be improved?
- **Resource Utilization:** Did you make the best use of available resources, like class notes, tutorials, or peer help? What additional resources could have been beneficial?
- **Time Management:** Reflect on how effectively you managed your time. What would you do differently to ensure a smoother process?

By considering these aspects, you will not only think about 'what' you would change but also 'how' and 'why,' which can be incredibly enlightening for future projects.

- Code Changes: I'll add sound effects, a timer, an entering page, and a menu with all the game things, I will make it like an actual game.
- Process Adjustments: I'll plan out everything before I start coding.
- Resource Utilization: I could use more CMU examples for ideas, I could have asked the teacher more questions.
- Time Management: I would start the movement and animation parts earlier.

Scroll down for submission checklist.

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Pre-Submission Checklist for Creative Tasks

Program Design

- [ ] Described what your program will do clearly.
- [ ] Incorporated at least two concepts from the current unit.
- [ ] Added annotations to explain colors, movements, and user interactions.

Canvas Design

- [ ] Sketched your canvas layout.
- [ ] Labeled any animation or user interaction on the canvas sketch.
- [ ] Detailed dominant colors and their importance.
- [ ] Described any animations or movements.

Code Design

- [ ] Listed at least four to six coding concepts or properties you plan to use.
- [ ] Explained why each concept or property is essential for your program.

Program Development and Coding

- [ ] Finished coding the program.
- [ ] Debugged and tested the program.
- [ ] Checked if the program meets the design criteria.

Difficulties and Opportunities

- [ ] Described a difficulty faced or an opportunity realized in detail.
- [ ] Explained the steps taken to resolve the difficulty or seize the opportunity.

Finished Project Code Outline

- [ ] Listed only one instance of each important coding concept used.
- [ ] Provided the line numbers for each coding concept.
- [ ] Included a brief definition for each coding concept.
- [ ] Highlighted the aspect you are most proud of.

Future Goals

- [ ] Discussed what specific changes you'd make if given more time.
- [ ] Reflected on your approach and planning for the project.
- [ ] Considered resource utilization and time management.

Final Check

- [ ] Revisited the rubric to make sure you meet all the criteria.
- [ ] Made sure your document is well-organized and free of grammatical errors.
- [ ] Ready to submit the project and this document.

By ticking off each item on this checklist, you ensure that you have met all the requirements and can confidently submit your work for evaluation.